ARTEMIdE—An Automated Underwater Material Recognition Method for Fluorescence LIDAR Invariant to Environmental Conditions

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***Abstract*— This article presents an automated underwater material recognition methodology for fluorescence light detection and ranging (LIDAR) invariant to environmental conditions (ARTEMIdE). Contrary to other state-of-the-art methods for submerged object recognition, ARTEMIdE can be applied when no *a priori* knowledge about environmental conditions is available and without resorting to any additional data besides the received signal and the fluorescence spectral signatures of the materials of interest. Experimental results over synthetic and real data show that ARTEMIdE is effective at automatically recognizing various object materials submerged at different depths within the water column. The presented approach reveals to provide great potential for many marine and submarine applications.**

***Index Terms*— Fluorescence light detection and ranging (LIDAR), invariance, LIDAR simulator, material discriminability, underwater object detection and recognition.**

I. Introduction

DETECTION and recognition of underwater objects and materials play an important role in many maritime and submarine applications, such as shipwreck detection [1], pipeline inspection [2], and undersea environmental monitoring [3]. An autonomous underwater vehicle (AUV) is generally sent to inspect the area of interest and acquire data with the multiple remote sensors of its payload. Suitable signal and image processing techniques need to be applied to properly retrieve the information of interest from the remotely sensed data. Generally, both laser and optical passive sensors (RGB or multi/hyperspectral) are mounted on-board the platform in order to gather both time-resolved waveforms and wavelength-resolved spectral data. This enables localization of underwater objects based on detection and ranging procedures as well as recognition of the detected materials by exploiting their optical properties. Within this framework, the use of fluorescence light detection and ranging (LIDAR) systems has attracted more and more interest [3]–[5], because both time-resolved and wavelength-resolved data acquisition capabilities are enclosed in a single compact system, without the need of resorting to multiple complementary sensors. In fact, a fluorescence LIDAR is basically a LIDAR with hyperspectral capabilities at the receiver. A sketch of fluorescence LIDAR phenomenology is provided in Fig. 1. Detection and ranging is performed searching for echoes within the elastic scattering (monochromatic) return waveform, generally induced with a green laser pulse, capable of deeply penetrating into the water column. Recognition of the underwater object material is then accomplished by suitably processing the spectral (polychromatic) return signal originated by inelastic scattering (fluorescence), extracted at the object range and collected at multiple wavelengths with fine spectral resolution by the system multichannel receiver. In fact, each specific material is characterized by unique fluorescence properties (depending on both material composition and excitation wavelength), i.e., by a fluorescence spectral signature that may be exploited for material recognition. Although other excitation wavelengths may be adopted [3], [5], fluorescence is generally induced with a laser pulse in the ultraviolet (UV) waveband, thus obtaining a signal spectrally distributed throughout most of the visible spectrum.

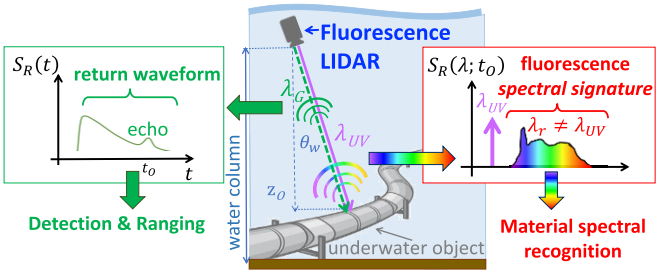


Fig. 1. Sketch of the fluorescence LIDAR phenomenology.

Over the years, fluorescence LIDAR systems have shown great potential for detecting and discriminating oil and chemical pollutants spilled or dissolved in the sea [6]–[10] as well as different algae species [5], [11], [12]. In these works, discrimination of different materials has been mostly carried out by visually inspecting the received signals [7]–[9], [11], [12] or tested using very simplified data sets [6], [10]. The problem at hand is indeed complicated. Not only the fluorescence signal returning from the underwater object is attenuated and spectrally colored by in-water propagation geometrical and optical effects but also the received signal includes the water volume backscattering contributions, i.e., the Raman

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signal and the fluorescence of sea water constituents [13], [14]. As far as we know, the only strategy so far capable of addressing the aforementioned issues in an automatic fashion is the one proposed in [4]. In [4], the received signal is compensated for water attenuation and inelastic scattering effects, and an estimate of the underwater object fluorescence spectral signature is then retrieved and used for spectral recognition of the materials of interest. This is achieved by exploiting the availability of LIDAR water-only (i.e., water free of any object) measurements in conjunction with a physics-based simplified signal model. However, water-only LIDAR measurements are seldom available in practice, and methodologies that resort to LIDAR signal simulators [5], [13] should be adopted to generate synthetic data related to the water signal contributions to be compensated for. Certainly, such an approach would require very precise knowledge of acquisition and environmental conditions as well as of LIDAR system parameters to accurately reproduce the water signal contributions. Whereas LIDAR parameters and viewing geometry are generally known because carefully planned before the mission, environmental conditions and, especially, water properties are hardly ever accurately predictable. Although some techniques exist to retrieve some of the water inherent optical properties, these may require complimentary sensors or complex LIDAR waveform inversion procedures that are still subject of on-going research [15], [16].

In this article, we propose an automated underwater material recognition methodology for fluorescence LIDAR that is invariant to water environmental conditions (ARTEMIdE). More specifically, ARTEMIdE is fully-automated and does not need any additional data besides the received signal and the fluorescence signatures of the materials of interest. Furthermore, by totally abstaining from any attempt to compensate for water effects, ARTEMIdE does not require accurate knowledge about environmental conditions. Drawing inspiration from a hyperspectral material detection method invariant to atmospheric conditions [17], ARTEMIdE makes use of an underwater fluorescence LIDAR signal model [13] and subspace projection concepts to build an invariant strategy that may be applied for fluorescence LIDAR material recognition in any submarine scenario in the presence of unknown environmental conditions. In addition, conventional and more robust subspace projection approaches are explored to be applied in this context. To the best of authors' knowledge, no such other underwater object recognition methodology has been reported previously.

Experimental results on both synthetic and real data reveal the great potential of the proposed methodology also when compared with the state-of-the-art automated underwater object recognition methods.

The structure of this article is as follows. ARTEMIdE design is detailed in Section II, experimental design and results are presented and discussed in Section III, and Section IV summarizes the main contributions of the work.

II. ARTEMIdE Design

Let us consider a fluorescence LIDAR system with a multichannel receiver collecting the incoming signal at d wavelengths $\{\lambda_r\}_{r=1}^{d}$. Throughout this article, we will use equivalently wavelength-dependent signal notation [e.g., $x(\lambda_r)$] and its vector-format counterpart (i.e., $\boldsymbol{x} = [x(\lambda_1), x(\lambda_2), \ldots, x(\lambda_d)]^T$, $\boldsymbol{x} \in \mathbb{R}^d$).

Goal of ARTEMIdE is to recognize a detected underwater object material with respect to a data base (DB) of object materials of interest, previously spectrally characterized by means of fluorescence measurements. We will consider a DB of N object materials, hereinafter denoted with the label set $\{O_i\}_{i=1}^{N}$. Each object material is associated with a single fluorescence spectral signature. As in [4] and [5], we make reference to the fluorescence spectral signatures in terms of fluorescence efficiency measured in $\mathrm{sr}^{-1} \cdot \mathrm{nm}^{-1}$, where sr stands for steradians. Fluorescence efficiency $\Phi_{O_i}(\lambda_{\mathrm{UV}}, \lambda_r)$ measures the fraction of laser energy impinging on the ith object at the UV excitation wavelength λ_{UV} that is re-emitted as fluorescence in the direction of the incoming laser beam, per unit wavelength and per unit solid angle, collected at the receiver wavelength λ_r [4], [5]. By collecting fluorescence efficiency measurements for each DB object material at each of the receiver wavelengths $\{\lambda_r\}_{r=1}^{d}$ of the LIDAR system, the DB is built as $\{\boldsymbol{\Phi}_{O_i}\}_{i=1}^{N}$, where $\boldsymbol{\Phi}_{O_i} \in \mathbb{R}^d$. Such spectral characterization may be carried out in laboratory, as described in [4] and [5].

Underwater object material spectral recognition is performed by ARTEMIdE only on the basis of the signal $\boldsymbol{S}_{R_O} = \boldsymbol{S}_R(t_O, \boldsymbol{\Phi}_O)$, which represents the received energy per unit wavelength at the output of the receiver detector [13]. Specifically, $\boldsymbol{S}_{R_O}$ collects, at each of the receiver wavelengths, the spectral signal (in photons $\cdot\, \mathrm{nm}^{-1}$) $S_R(t_O, \lambda_r, \Phi_O(\lambda_{\mathrm{UV}}, \lambda_r))$ received by the LIDAR system from the (unknown) detected object O, extracted at the object range $r_O = (2/v)t_O = (z_O/\cos\theta_w)$. Here, v is the laser beam velocity in water, z_O is the object depth, and θ_w is the in-water laser beam propagation angle, as illustrated in the sketch of Fig. 1. Specifically, based on $\boldsymbol{S}_{R_O}$, the goal is to decide if the detected object material O corresponds to one of the DB object materials $\{O_i\}_{i=1}^{N}$ and, if so, to which DB material, it most likely corresponds. This may be cast in terms of multiple $(N+1)$ hypotheses testing [18], as follows:

$$\begin{aligned} &H_0 : O \notin \{O_i\}_{i=1}^{N} \\ &\{H_i : O = O_i\}_{i=1}^{N}. \end{aligned} \tag{1}$$

Fulfillment of the H_i hypothesis indicates that the detected object material is recognized as O_i, whereas rejection of all $\{H_i\}_{i=1}^{N}$ hypotheses in favor of the null hypothesis H_0 means that the detected object material is not recognized as any of the materials in the DB.

As said in the Introduction, the fluorescence signature $\boldsymbol{\Phi}_O$ of the detected object is not directly observed. Rather, it is included within $\boldsymbol{S}_{R_O}$ [13, eqs. (1)–(15)], transformed by in-water propagation effects and intertwined with other received signal contributions. Therefore, to develop an automated and invariant methodology based on the fluorescence signature, an accurate underwater fluorescence LIDAR signal model needs to be invoked. ARTEMIdE lays its foundations over the underwater fluorescence LIDAR signal model and simulator [13] outlined in Section II-A.

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A. Underwater Fluorescence LIDAR Signal Model and Simulator

The signal model employed in this work is the one proposed in [13] and employed in "A FluoRescence Lidar Simulator for Underwater Object DeTEction Applications" (AFRODiTE) [13]. Specifically, AFRODiTE models the inelastic backscattering signals induced by an UV laser pulse as a result of the laser beam interaction with the water medium, an underwater object, and (if reached) the sea bottom.

In this work, we restrict our attention to submarine scenarios and assume underwater objects larger than the transmitted laser beamwidth at the object depth. Under these assumptions, the equation ruling the total received signal $S_R(t,\lambda_r)$ (in photons$\cdot$nm^{-1}) collected by the LIDAR detector at time t and wavelength λ_r is as follows:

$$S_R(t,\lambda_r) = S_O(t,\lambda_r)+S_W(t,\lambda_r)-S_{\rm sh}(t,\lambda_r)+S_n(t,\lambda_r) \quad (2)$$

where $S_O(t,\lambda_r)$ is the signal returned by the underwater object material, and $S_W(t,\lambda_r) - S_{\rm sh}(t,\lambda_r)$ is the signal returned by the water column. $S_{\rm sh}(t,\lambda_r)$ accounts for the loss of backscattering signal with respect to the signal $S_W(t,\lambda_r)$ that would be received from the water column free of the object (which, instead, "casts a shadow" behind itself) and $S_n(t,\lambda_r)$ is the shot noise term [13], [19]. The fluorescence signature $\Phi_O(\lambda_{\rm UV},\lambda_r)$ of the underwater object contributes to $S_O(t,\lambda_r)$ together with the downward and upward water transmittances and other quantities depending on the system and acquisition geometry. The term $S_W(t,\lambda_r)-S_{\rm sh}(t,\lambda_r)$ includes the Raman signal, due to an inelastic scattering phenomenon different from fluorescence [13] ([14, Section 5.1.4]), and the water fluorescence contributions pertinent to the in-water path, combined likewise with system- and geometry-related quantities [13]. Each term of equation (2) is obtained by following a channel-modeling approach, where the laser pulse is convolved with the impulse responses that model the underwater object, the entire water column, and the water column portion shadowed by the object itself [13]. A fully detailed expansion of each term of equation (2) can be found in [13] and [19].

The simulator is capable of reproducing the received signal $S_R(t,\lambda_r)$ on the basis of user-specified parameters inherent to LIDAR system, acquisition geometry, environmental conditions, as well as of the object fluorescence spectral signature $\Phi_O(\lambda_{\rm UV},\lambda_r)$. A block diagram of the simulator is sketched in Fig. 2, which shows some further details of the model implemented in the simulator. In the following, we will indicate with $\tilde{S}_R(t,\lambda_r)$ the synthetic signal generated by the simulator to reproduce $S_R(t,\lambda_r)$ of equation (2) in the presence of an underwater object characterized by fluorescence signature $\Phi_O(\lambda_{\rm UV},\lambda_r)$. The corresponding wavelength-dependent signal $\tilde{S}_R(t_O,\lambda_r)$ extracted at the detected object range given by t_O can also be expressed as a d-dimensional vector in $\mathbb{R}^d$ and will be indicated as $\tilde{\boldsymbol{S}}_{R_O}$.

B. Signature Spaces Generation

In this work, we assume that LIDAR system characteristics and acquisition geometry parameters are known and we restrict our study to design a recognition methodology invariant to environmental conditions. There is no loss of generality in this assumption, which can be relaxed at no cost in the case of less accurately known system and geometry parameters.

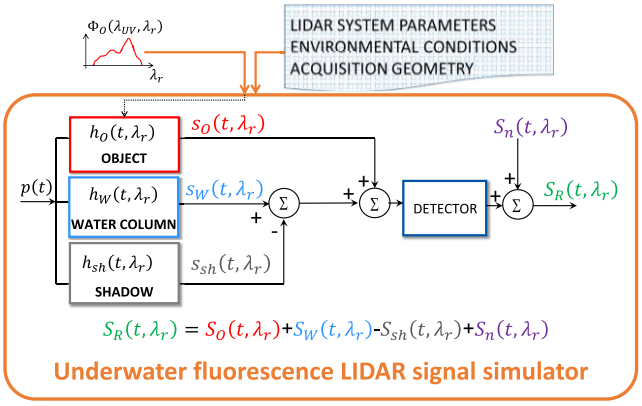


Fig. 2. Block diagram of the underwater fluorescence LIDAR simulator employed. A channel-modeling approach is adopted — the laser pulse $p(t)$ is convolved with the impulse responses modeling the underwater object, the entire water column, and the water column portion shadowed by the object itself. A "gateable" detector allowing nanosecond time resolution is also implemented in the simulator. The simulator is fed with parameters inherent to LIDAR system, acquisition geometry, and environmental conditions as well as with the underwater object fluorescence spectral signature.

The idea behind ARTEMIdE is given as follows. For each of the N DB object materials, a set of D synthetic received signals (or, equivalently, d-dimensional vectors) is generated. Each signal is obtained for one of the possible D configurations of the parameters characterizing the environmental conditions

$$\left\{ O_i : \Psi_i = \left\{ \boldsymbol{\psi}_i^{(j)} = \tilde{\boldsymbol{S}}_{R_{O_i}}^{(j)} \right\}_{j=1}^{D} \right\}_{i=1}^{N} \quad (3)$$

where $\boldsymbol{\psi}_i^{(j)} = \tilde{\boldsymbol{S}}_{R_{O_i}}^{(j)}$, detailed at the bottom of page,[1] reproduces the signal received by the LIDAR system from the ith submerged object O_i in the jth environmental condition extracted at the range given by t_O. The set of spectra Ψ_i, also called elsewhere signature space [19], represents all the possible ways the signal returned from the submerged object O_i and received by the LIDAR system might look like in the various different D environmental conditions. The name signature space refers to the fact that Ψ_i can be seen as a cloud of D data points in $\mathbb{R}^d$ originated by the single data point $\boldsymbol{\Phi}_{O_i}$ and by the uncertainty about environmental conditions.

The generic signature space Ψ_i is generated by multiple applications of the simulator previously described, where the simulator is fed with $\boldsymbol{\Phi}_{O_i}$ and with multiple variable environmental parameters. Here, the environmental variability should be large enough to capture the reasonable expected environmental conditions affecting the test signal and can be properly tuned based on the specific scenario and knowledge degree.

[1] $\boldsymbol{\psi}_i^{(j)} = \tilde{\boldsymbol{S}}_{R_{O_i}}^{(j)} = [\tilde{S}_R^{(j)}(t_O,\lambda_1,\Phi_{O_i}(\lambda_{\rm UV},\lambda_1)), \tilde{S}_R^{(j)}(t_O,\lambda_2,\Phi_{O_i}(\lambda_{\rm UV},\lambda_2)), \ldots, \tilde{S}_R^{(j)}(t_O,\lambda_d,\Phi_{O_i}(\lambda_{\rm UV},\lambda_d))]^T$

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C. Subspace Projection

Once the set of N signature spaces $\{\Psi_i\}_{i=1}^N$ is generated, the test signal returning from the unknown detected object O is projected onto the subspaces derived by each signature space. Subspace projection procedures will be detailed in Sections II-C1 and II-C2. The decision rule adopted is then to recognize the detected object as the DB object material O_k (fulfillment of H_k), $k \in \{1, 2, \ldots, N\}$, corresponding to the signature subspace for which the projection error is the lowest, as in [17], provided that it exceeds a suitable threshold η_{H_0}. That is,

$$H_k : \begin{cases} k = \arg\min_{i=1,2,\ldots,N} \left\{\varepsilon_{\Psi_i}^{(O)}\right\} \\ \varepsilon_{\Psi_k}^{(O)} > \eta_{H_0} \end{cases} \tag{4}$$

where $\varepsilon_{\Psi_i}^{(O)}$ is the projection error of the test signal of object O onto the subspace associated with Ψ_i and will be detailed below.

Since the error in approximating the test signal with its projection onto the subspace depends on the amplitude of the test signal itself, we actually project the amplitude-normalized test vector $\boldsymbol{S}'_{R_O} = (\boldsymbol{S}_{R_O}/\|\boldsymbol{S}_{R_O}\|)$, as in [17]. In this regard, ARTEMIdE is also invariant to the magnitude of the object material fluorescence signature, similarly to other state-of-the-art recognition methodologies [4]. The projected vector will be denoted with $\hat{\boldsymbol{S}}_{R_O}^{(\Psi_i)}$ and each of the projection errors in (4) is evaluated as $\varepsilon_{O_i} = \|\boldsymbol{S}'_{R_O} - \hat{\boldsymbol{S}}_{R_O}^{(\Psi_i)}\|$.

Two approaches may be considered for subspace projection depending on the subspace generation process, namely, unconstrained subspace projection (USP) and constrained subspace projection (CSP).

1) USP: The most straightforward way to proceed when dealing with large sets of data, like the signature space Ψ_i, is searching for a low-dimensional linear model capable of providing an accurate representation of the various spectra of Ψ_i. A possible approach is to find an orthonormal basis, collected into the $d \times K_i$ matrix $\boldsymbol{M}_i = [\boldsymbol{m}_{i,1}, \boldsymbol{m}_{i,2}, \ldots, \boldsymbol{m}_{i,K_i}]$, spanning a K_i-dimensional subspace, for instance by means of the singular value decomposition (SVD). The projected test signal is then expressed as a linear combination of the basis vectors $\hat{\boldsymbol{S}}_{R_O}^{(\Psi_i)} = \boldsymbol{M}_i \boldsymbol{a}_{\boldsymbol{s}}^{(i)} = \sum_{q=1}^{K_i} a_{sq}^{(i)} \boldsymbol{m}_{i,q}$, with $\boldsymbol{a}_{\boldsymbol{s}}^{(i)}$ indicating the $K_i \times 1$ vector of coefficients. The coefficients are derived by the following least-square problem with closed-form solution as follows:

$$\boldsymbol{a}_{\boldsymbol{s}}^{(i)} = \arg\min_{\boldsymbol{a}} \left\{ \left\| \boldsymbol{S}'_{R_O} - \boldsymbol{M}_i \boldsymbol{a} \right\| \right\} = \boldsymbol{M}_i^{\#} \boldsymbol{S}'_{R_O} \tag{5}$$

where $\boldsymbol{M}_i^{\#} = (\boldsymbol{M}_i^T \boldsymbol{M}_i)\boldsymbol{M}_i^T$ is the pseudo-inverse matrix of $\boldsymbol{M}_i$.

Although very commonly employed [17], [20], there are two main limitations deriving from this approach. The first is how to determine the "optimal" subspace rank K_i. With a too low K_i, some important information crucial for recognition may be left out, whereas a too high K_i is likely to include too much (and undesired) variability. One general criterion is to choose the K_i providing a low enough average normalized approximation error ϵ_{Ψ_i} on the signature space [17]

$$\epsilon_{\Psi_i} = \frac{\sum_{j=1}^{D} \|\boldsymbol{\psi}_i^{(j)} - \sum_{q=1}^{K_i} a_{sq}^{(i)} \boldsymbol{m}_{i,q}\|^2}{\sum_{j=1}^{D} \|\boldsymbol{\psi}_i^{(j)}\|^2}. \tag{6}$$

The second limitation is that without imposing any constraint onto the coefficients, the size (which is different from the dimension K_i) of the subspace is not constrained. For instance, for $K_i = 3$, the spanned subspace has an extent as the entire $\mathbb{R}^3$. Thus, the resulting subspace is likely to accommodate also spectra much different from that of the test object [21]. The use of constrained subspaces, explored in other contexts [21]–[23], overcomes both limitations.

2) CSP: By denoting with $\boldsymbol{\Psi}_i = [\boldsymbol{\psi}_i^{(1)}, \boldsymbol{\psi}_i^{(2)}, \ldots, \boldsymbol{\psi}_i^{(D)}]$ the $d \times D$ matrix collecting, as columns, the Ψ_i signature space spectra, we do not estimate a basis for the subspace but let the $\{\boldsymbol{\psi}_i^{(j)}\}_{j=1}^D$ span the subspace. However, in doing so, we allow only for linear combinations with nonnegative coefficients of the $\{\boldsymbol{\psi}_i^{(j)}\}_{j=1}^D$, i.e., $\hat{\boldsymbol{S}}_{R_O}^{(\Psi_i)} = \boldsymbol{\Psi}_i \boldsymbol{a}_{\boldsymbol{s}}^{(i)}$ with

$$\boldsymbol{a}_{\boldsymbol{s}}^{(i)} = \arg\min_{\boldsymbol{a} \geq \boldsymbol{0}} \left\{ \left\| \boldsymbol{S}'_{R_O} - \boldsymbol{\Psi}_i \boldsymbol{a} \right\| \right\}. \tag{7}$$

This nonnegative least-squares problem can be solved numerically. Geometrically, a nonnegatively constrained subspace is a d-dimensional polyhedral cone with D edges [22], and this also makes sense physically because we do not admit unphysical combinations of spectra (e.g., leading to a negative spectrum).

D. ARTEMIdE Overall Scheme

In summary, ARTEMIdE is made up of three phases, namely: 1) signature spaces generation; 2) subspace projection; and 3) object material recognition. The overall ARTEMIdE scheme is displayed in Fig. 3.

III. Experimental Results

Two sets of experiments were performed. In the first experiment, we used synthetic data in order to thoroughly test ARTEMIdE in a controlled environment. Very preliminary results with synthetic data were shown in [34]. Here, more extensive experiments were performed. Furthermore, USP and CSP were compared to choose which is the most effective subspace projection approach to be employed in ARTEMIdE. Real data were used in the second set of experiments, where ARTEMIdE was compared to other state-of-the-art underwater material recognition methodologies.

A. Experiments With Synthetic Test Data

1) Experimental Design:

a) Synthetic test data: A set of synthetic test data was obtained by simulating $N = 8$ object materials as submerged 10 m deep in optically complex Case 2 [24] waters. A LIDAR system with the characteristics reported in Table I was simulated as arranged below the water surface and pointing downward with an in-water propagation angle of $\theta_w = 5°$. The simulated acquisition geometry is depicted in Fig. 4. The spectral signatures of the eight object materials are shown

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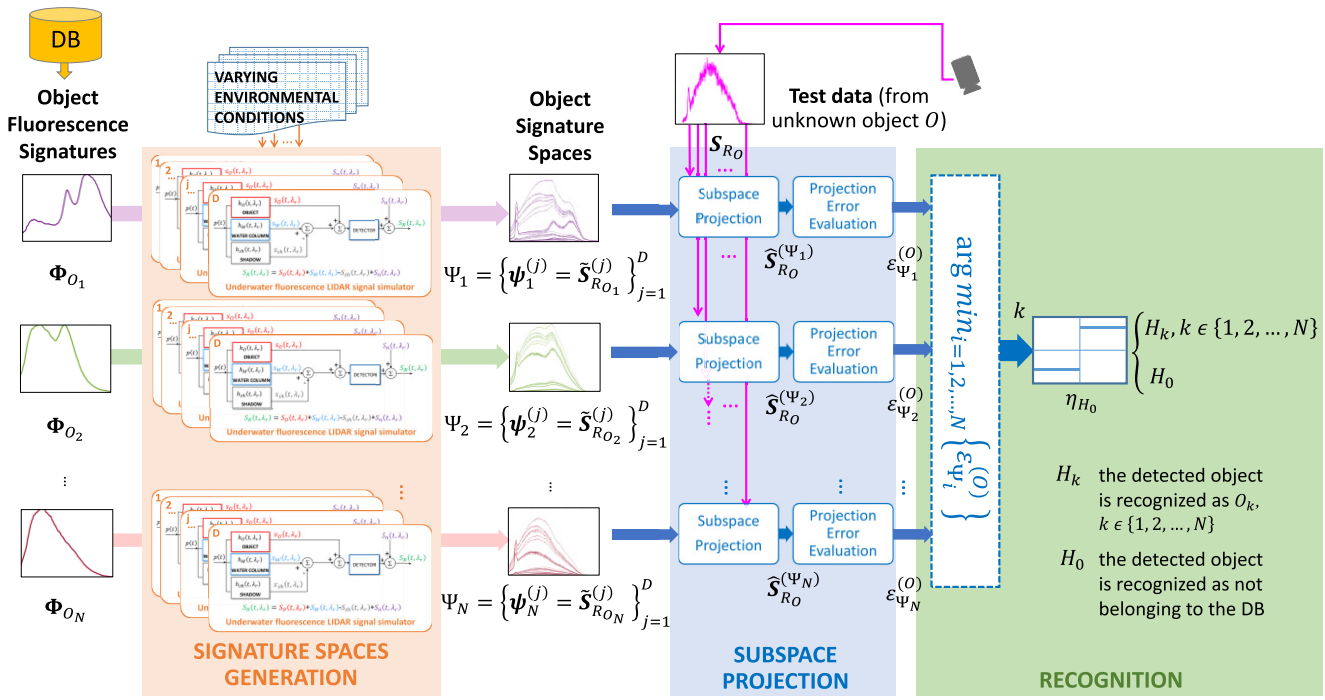


Fig. 3. ARTEMIdE overall scheme.

TABLE I
LIDAR SYSTEM PARAMETERS FOR SYNTHETIC DATA SET EXPERIMENTS

Excitation wavelength [nm]	355
Pulse energy [mJ]	50
Pulse FWHM [ns]	5
TX laser beam div. [mrad]	5
Emission efficiency	0.8
Telescope diameter [cm]	25
Receiver optical efficiency	0.5
Receiver field of view [sr]	1.1 10^{-3}
RX spectral range [nm]	[380, 650]
RX spectral resolution [nm]	4
Detector size [mm]	75
Detector focal length [mm]	300
Detector quantum efficiency	0.75
Gate duration [ns]	5
Gating delay [ns]	1

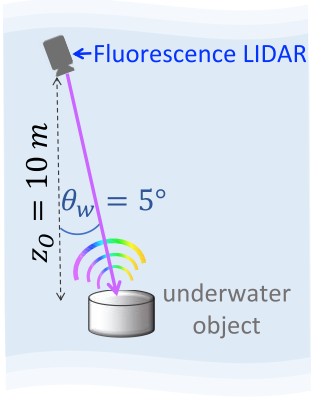


Fig. 4. Simulated viewing geometry for the synthetic test data.

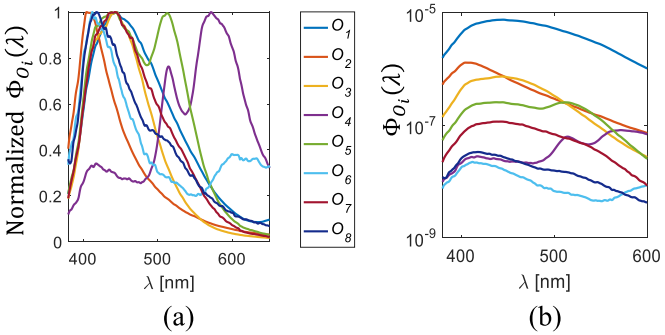


Fig. 5. Fluorescence spectral signatures of the eight object materials. They also served as spectral DB with respect to which perform spectral recognition. To increase readability, (a) spectra are normalized within [0, 1] and (b) spectra are plotted in the Y-logarithmic scale.

in Fig. 5, whereas Table II provides a list of the specific object materials employed. They mostly refer to man-made materials covered with different paint coatings. In order to improve the visibility of the weakest signals with respect to the strongest ones, normalized data are shown in Fig. 5(a), whereas a Y-logarithmic scale is adopted in Fig. 5(b) to show nonnormalized spectra. As is evident in Fig. 5(a), each object material is characterized by specific fluorescence spectral features, but several materials have very similar fluorescence properties and discrimination of such materials might be difficult. Fig. 5(b) also shows that some object materials fluoresce more than others. Clearly, object materials with more peculiar spectral features (e.g., O_4 and O_5) or characterized by stronger fluorescence (e.g., O_1) are expected

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TABLE II
DB OBJECT MATERIALS FOR SYNTHETIC DATA SET EXPERIMENTS

O_1	Fiberglass
O_2	Neoprene
O_3	Blue Tin
O_4	Alpine Green Aluminum
O_5	Orange Aluminum
O_6	Opaque Orange Aluminum
O_7	Dark Green Aluminum
O_8	Greyish Green Aluminum

TABLE III
OPTICALLY ACTIVE SUBSTANCES SETTINGS FOR SYNTHETIC TEST DATA

Parameter	$C[mg/m^3]$	$X\ [m^{-1}]$	$Y\ [m^{-1}]$
	$17.90 \cdot 10^{-2}$	$6.28 \cdot 10^{-2}$	$1.34 \cdot 10^{-2}$
μ_{Log}	$Log(C)$	$Log(X)$	$Log(Y)$
	-0.7471	-0.3737	-0.8729
σ_{Log}	$\sigma_{Log(C)}$	$\sigma_{Log(X)}$	$\sigma_{Log(Y)}$
	0.15	0.20	0.15
ρ	$\rho_{Log(C)Log(X)}$		$\rho_{Log(C)Log(Y)}$
	0.40		0.40

TABLE IV
FA AND HA SETTINGS FOR SYNTHETIC TEST DATA

Acid	HA	FA
Type	HA6	FA7
Concentration	$Y_{HA}\ [g/m^3]$	$Y_{FA}\ [g/m^3]$
μ	$2.90 \cdot 10^{-2}$	$2.23 \cdot 10^{-1}$
σ	$5.0 \cdot 10^{-3}$	$5.0 \cdot 10^{-2}$

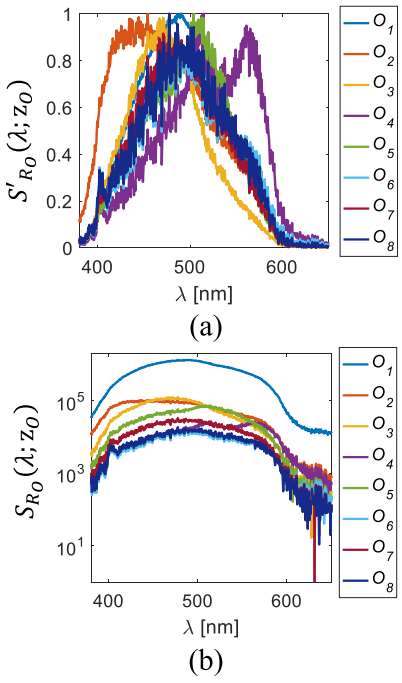


Fig. 6. Simulated test data for the eight submerged object materials. One random realization is shown for each object material. (a) Normalized data. (b) Nonnormalized data—Y-logarithmic scale is employed to increase the readability.

to be more easily recognized, because of their more distinctive or stronger, respectively, return signals. Conversely, object materials with low fluorescence and relatively flat fluorescence spectrum (e.g., O_6, O_7, O_8) are likely to lead to worse recognition performance. Synthetic test data were generated with AFRODiTE simulator [13] according to the model in (1). A number of $N_r = 100$ signal realizations for each object material was generated for the test data by randomly drawing the corresponding environmental parameters from suitable probability distribution functions (PDFs). Specifically, the concentrations of phytoplankton (expressed in terms of C, the concentration of chlorophyll a and pheophytin a, in $\text{mg} \cdot \text{m}^{-3}$), nonchlorophyllous particles (expressed in terms of X, the scattering coefficient of nonchlorophyllous particles at 550 nm, in m^{-1}), and yellow substance (expressed in terms of Y, the absorption coefficient of yellow substance at 440 nm) were drawn from suitable log-normal PDFs. The log-normal distribution has in fact been found to well represent the behavior of bio-optical variables such as those associated with the optically active substances of seawater [25]. The corresponding parameters are reported in Table III. Specifically, Table III reports the mean value μ_{Log} and standard deviation σ_{Log} of the logarithm of C, X, and Y together with the correlation coefficients between the pairs of variables $\text{Log}(C), \text{Log}(X)$, and $\text{Log}(C), \text{Log}(Y)$ [26], [27]. Table IV reports the types of humic acid (HA) and fulvic acids (FA) and concentrations thereof (Y_{HA} and Y_{FA}, in $\text{g} \cdot \text{m}^{-3}$) employed to simulate fluorescence backscattering by yellow substance [13], [28]. Specifically, yellow substance fluorescence was modeled by employing the acid types denoted by HA6 and FA7 in [28] together with acid concentrations normally distributed around the parameters tabled in [29].

The tunable signal-to-noise ratio (SNR) model proposed in [18] was employed for test data generation. Here, the parameter tuning the signal-dependent noise variance was set by specifying the loss $L|\text{dB}$ (in decibels) in SNR with respect to the average SNR obtained when the tuning parameter is equal to 1 [19]. The reader is referred to [19] for specific details about the noise model, but, basically, the higher the $L|\text{dB}$, the lower the SNR, the worse the noise conditions. Here, the test data were simulated with $L|\text{dB} = 40$. One random realization of the test data simulated for each of the eight submerged object materials is shown in Fig. 6(a) and (b). To improve the visibility of the weakest signals with respect to the strongest ones, normalized data are shown in Fig. 6(a), whereas a Y-logarithmic scale is adopted in Fig. 6(b) to show nonnormalized data. Clearly, object materials with stronger fluorescence properties led to stronger return signals [Fig. 6(b)]. As shown in Fig. 6(a), although the signals from the submerged object materials are combined with the water

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TABLE V
ENVIRONMENTAL PARAMETERS EMPLOYED FOR SIGNATURE SPACE GENERATION FOR SYNTHETIC DATA SET EXPERIMENTS

$C\ [mg/m^3]$	$X\ [m^{-1}]$	$Y\ [m^{-1}]$	HA type	FA type	$Y_{HA}\ [g/m^3]$	$Y_{FA}\ [g/m^3]$
$12.30 \cdot 10^{-2}$	$5.60 \cdot 10^{-2}$	$1.61 \cdot 10^{-2}$	HA10	FA11	$9.00 \cdot 10^{-3}$	$5.80 \cdot 10^{-2}$
$39.40 \cdot 10^{-2}$	$1.43 \cdot 10^{-2}$	$4.11 \cdot 10^{-2}$	HA10	FA9	$1.40 \cdot 10^{-2}$	$2.34 \cdot 10^{-1}$
$25.10 \cdot 10^{-2}$	$5.01 \cdot 10^{-2}$	$1.59 \cdot 10^{-2}$	HA8	FA8	$1.14 \cdot 10^{-1}$	$5.50 \cdot 10^{-1}$
$61.80 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$	$2.49 \cdot 10^{-2}$	HA4	FA8	$2.60 \cdot 10^{-2}$	$7.24 \cdot 10^{-1}$

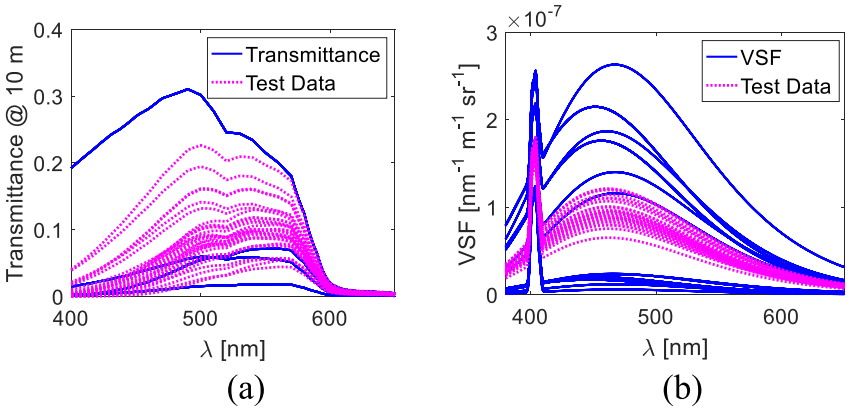


Fig. 7. (a) Transmittances at 10 m for the four triplets of water optically active substance parameters reported in Table V. (b) VSFs for the 16 combinations of acid types and concentrations reported in Table V. Transmittances and VSFs for 30 random realizations of the test data are superimposed in pink dotted curve.

column backscattering, some spectral features, especially the most distinctive ones, are still clearly visible in the spectra.

b) Signature space generation: A spectral DB composed by the same eight object material spectra as those employed to simulate the test data and shown in Fig. 5 was employed for spectral recognition.

For each object material, a signature space was generated by varying the environmental parameters as shown in Table V, while keeping the same acquisition geometry and LIDAR system parameters as in the test data. More specifically, signature spaces were generated by reproducing 64 environmental conditions by allowing for (see Table V) four configurations for the triplets (C, X, Y), four configurations for the HA and FA acid type pairs, and four configurations for the $(Y_{\mathrm{HA}}, Y_{\mathrm{FA}})$ pairs. The values in Table V were taken from tabled values in the literature and refer to both clearer Case 1 and more optically complex Case 2 waters [13], [24], [26], [28], [29].

The transmittances evaluated at $z_O = 10$ m for the four (C, X, Y) triplets are shown in Fig. 7(a), whereas the volume inelastic scattering functions (VSFs) for the 16 combinations of acid types and concentrations are shown in Fig. 7(b). The pink dotted curves in Fig. 7(b) are transmittances and VSFs employed to generate 30 random realizations of the test data (as described in Section III-A1a and with parameters as in Tables III and IV). As is evident, the range of variation of the curves for the test data is well captured by the high variability of the reproduced environmental conditions, which span from clear blue waters to turbid and highly fluorescent waters.

For actual signature space generation, we decided to introduce further variability by varying the aforementioned $L|\mathrm{dB}$ noise parameter. Specifically, we let it take three values, i.e., $L|\mathrm{dB} = \{20, 30, 50\}$. This led to signature spaces including $D = 64 * 3 = 192$ spectra. Two examples of signatures spaces for O_4 and O_5 object materials (i.e., Ψ_4 and Ψ_5) are shown in Fig. 8(a) and (b), displayed in Y-logarithmic scale to improve readability. A number of ten random realizations of the test data for each object material are superimposed in pink for convenience. As is evident, the test data are well encapsulated within the wide variability addressed by the signature spaces. More specifically, the test data place themselves in the "region" of the weakest signals of the signature spaces, as is reasonable to expect being the signature spaces generated for both Case 1 and Case 2 waters and the data simulated for Case 2 waters.

c) Subspace projection: For USP, results will be discussed for a choice of $\{K_i\}_{i=1}^{N=8} = K = 5$, which was shown to provide low enough (i.e., lower than 10^{-5}) average normalized approximation errors $\{\epsilon_{\Psi_i}\}_{i=1}^{N=8}$ on the signature spaces for all eight object materials. Plot of the $\{\epsilon_{\Psi_i}\}_{i=1}^{N=8}$ with respect to the number of subspace components is shown in Fig. 9. Nonetheless, results will be inspected for a much wider range of K values ($K \in [3, 100]$), thus investigating the dependence of USP performance on the number of subspace components. For CSP no specific parameter setting was required.

d) Performance evaluation measures: Prior examining actual recognition performance measures, intermediate results will be inspected in terms of projection errors $\{\varepsilon_{\Psi_i}^{(O_u)}\}_{\substack{u=1,\ldots,N \\ i=1,\ldots,N}}$ obtained, with both USP and CSP, by projecting the N_r

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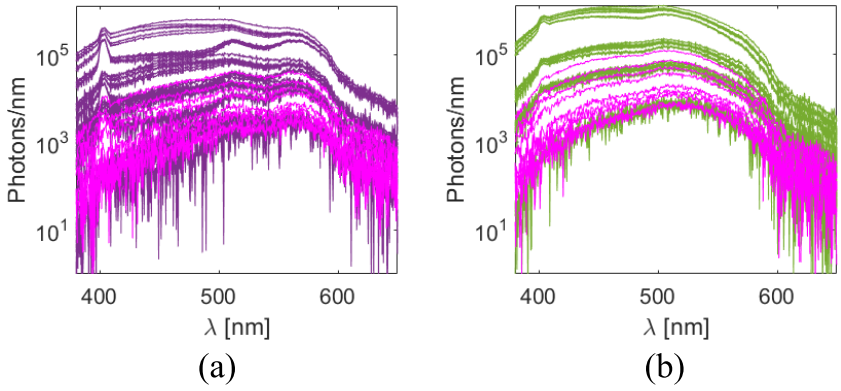


Fig. 8. Signature spaces (a) Ψ_4 (for O_4) and (b) Ψ_5 (for O_5). The signature spaces were obtained with the combination of the parameters in Table V. A number of ten random realizations of the test data for (a) O_4 and (b) O_5 are superimposed in pink. The graphs are in the Y-logarithmic scale to improve the readability.

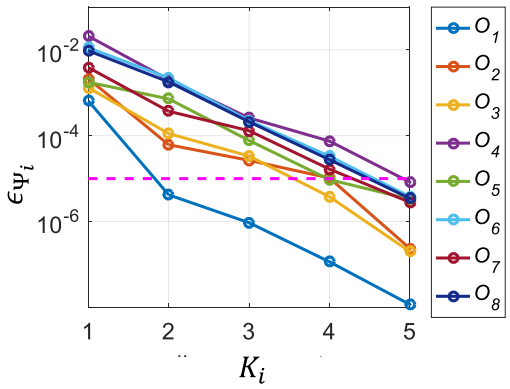


Fig. 9. Average normalized approximation errors on the signature spaces for the eight object materials plotted with respect to the number of subspace components. The x-axis has been cropped to the number of components providing an error lower than 10^{-5} for all object materials (i.e., $\{K_i\}_{i=1}^{N=8} = K = 5$).

realizations of the test signals for the eight object materials over the generated signature spaces.

Then, ARTEMIdE recognition performance will be assessed by evaluating confusion matrix-derived metrics typically employed in multiclass classification tasks [30]. Specifically, indicating with v_{ij} the number of realizations of the test data for object material O_i that are recognized as corresponding to object material O_j, the $N \times N$ matrix $\boldsymbol{V}$ formed by the v_{ij} elements is commonly known as confusion matrix [30]. The diagonal contains the number of test data realizations correctly recognized. Clearly, the closer to diagonal $\boldsymbol{V}$ is, the better is the performance.

From the confusion matrix, a per-object accuracy $A_i\%$, expressed in percentage, can be derived as follows [30]:

$$\left\{ A_i\% = \frac{v_{ii}}{N_r} \cdot 100 \right\}_{i=1}^{N}. \tag{8}$$

The $A_i\%$ can be regarded as the probability (in percentage) of correct recognition for object material O_i.

An overall accuracy measure OA%, expressed in percentage, can be evaluated by averaging the $A_i\%$ over the number of object materials [30]

$$\text{OA}\% = \frac{1}{N} \sum_{i=1}^{N} A_i\%. \tag{9}$$

2) Results Discussion: Fig. 10 shows a box-and-whiskers plot of the projection errors for both USP applied with $K = 5$ (in blue) and CSP (in purple). This plot gives a synoptic view of the projection error distributions. From the plot, it is evident that the CSP provided more robust results than USP, exhibiting in most cases a much greater separation between $\varepsilon_{\Psi_u}^{(O_u)}$ values (which should be the smallest) and the other $\{\varepsilon_{\Psi_i}^{(O_u)}\}_{i \neq u}$. The plots also reveal that objects O_1, O_2, O_3, O_4, O_5 should provide better recognition performance than the others, especially for the CSP approach. For these objects, in fact, the separation between $\varepsilon_{\Psi_u}^{(O_u)}$ and $\{\varepsilon_{\Psi_i}^{(O_u)}\}_{i \neq u}$ is much more evident than for the other objects.

Fig. 11(a) and (b) shows the confusion matrices obtained for USP applied with $K = 5$ and for CSP, respectively. Both matrices exhibit a nearly-diagonal structure with most of off-diagonal elements equal to zero or assuming low values. The confusion matrix for CSP exhibits a closer-to-diagonal structure than that for USP, which instead reveals a higher occurrence of both low diagonal elements and nonnull or high off-diagonal elements.

More quantitative information can be derived from the per-object accuracy graphs plotted in Fig. 11(c) and (d) for USP ($K = 5$) and CSP, respectively. The OA% is also displayed on both graphs. As is evident, even though no *a priori* information (in any form) about the environmental conditions affecting the test data is employed, ARTEMIdE provides a rather good submerged material discrimination ability. Namely, USP provides an OA% of about 70% while CSP nearly reaches an OA% of 80%. The graphs reveal that recognition performance is much better for CSP application. In fact, six objects ($\{O_1, O_2, O_3, O_4, O_5, O_6\}$) for CSP versus four objects ($\{O_1, O_3, O_4, O_5\}$) for USP were recognized with good accuracy ($A_i\% > 70$). CSP managed to recognize with high accuracy also O_6, characterized by low fluorescence ad without peculiar spectral features. In general, recognition performance for the low fluorescent $\{O_7, O_8\}$ was much worse ($A_i\% < 50$) for both USP and CSP—especially for O_7,

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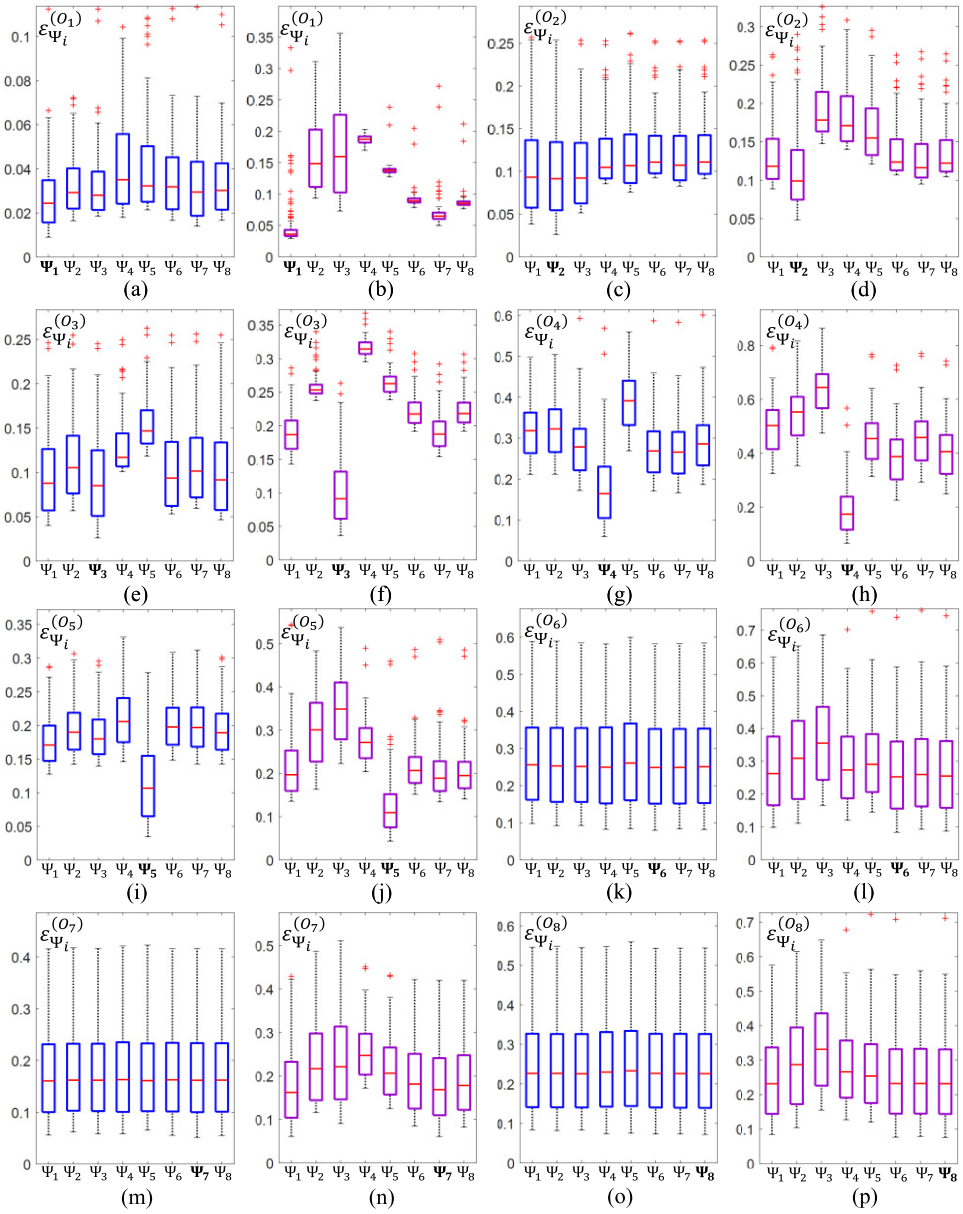


Fig. 10. Box-and-whiskers plots of projection errors $\{\varepsilon_{\Psi_i}^{(O_u)}\}_{\substack{u=1,\ldots,N\\ i=1,\ldots,N}}$ for the N_r realizations of the test signals for each of the $\{O_u\}_{u=1}^{N}$ tested object materials over the $\{\Psi_i\}_{i=1}^{N}$ subspaces for both USP (in blue) and CSP (in purple). Each box spans from the 25% percentile to the 75% percentile (and, thus, its extent is equal to the interquartile range—IQR) of the corresponding data. The median of the data is plotted as a horizontal red line within the box. In this plot, the two whiskers emanating from the box edges extend, in each direction, to the most extreme data value that is not considered as an outlier. Data distant more than 1.5 times the IQR (i.e., the box extent) from the corresponding box edges are considered outliers and are plotted with red cross markers. (a) and (b) $\varepsilon_{\Psi_i}^{(O_1)}$. (c) and (d) $\varepsilon_{\Psi_i}^{(O_2)}$. (e) and (f) $\varepsilon_{\Psi_i}^{(O_3)}$. (g) and (h) $\varepsilon_{\Psi_i}^{(O_4)}$. (i) and (j) $\varepsilon_{\Psi_i}^{(O_5)}$. (k) and (l) $\varepsilon_{\Psi_i}^{(O_6)}$. (m) and (n) $\varepsilon_{\Psi_i}^{(O_7)}$. (o) and (p) $\varepsilon_{\Psi_i}^{(O_8)}$. (a), (c), (e), (g), (i), (k), (m), and (o) USP. (b), (d), (g), (h), (j), (n), (p) CSP.

which was mostly confused with O_1, as clearly shown by the confusion matrices. The two objects have indeed a very similar fluorescence spectral character [see Fig. 5(a)], although O_1 has a much stronger fluorescence signature making its signal stronger and more easily recognizable.

Fig. 12 plots OA% with respect to K. Fig. 12 clearly shows that USP performance strongly depends on K. For the smallest $K = 3$ value tested, OA% is slightly above 50, then it reaches its maximum values (OA% $\sim$ 70) for $K \in [5, 10]$, and then steeply decreases to very low values for increasing K. This trend was expected because too few subspace components may leave out the spectral information that is crucial for recognition, whereas too many components are likely to include information not essential to recognition that contribute to an overall decrease in recognition capabilities. In Fig. 12, the OA% obtained with the CSP is shown for comparison in purple dashed and is always higher than the OA% obtained with different values of K for USP. Hence, the CSP revealed to be the most-effective and practical subspace projection approach to be applied within ARTEMIdE. It does not require any parameter setting, and it manages to effectively capture the variability essential for recognition. In the following, CSP

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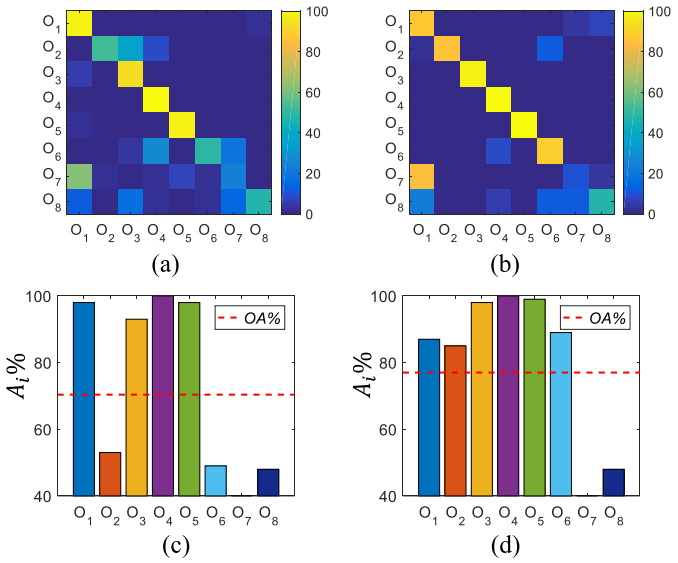


Fig. 11. Spectral recognition performance in terms of (a) and (b) confusion matrices and (c) and (d) per-object accuracies A_i% for (a) and (c) USP applied with $K = 5$ and (b) and (d) CSP. (c) and (d) Overall accuracy OA% is superimposed as a red dashed curve for convenience.

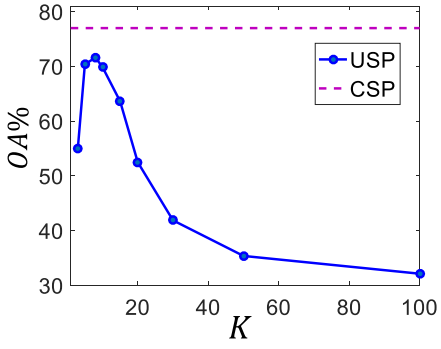


Fig. 12. Overall accuracy OA% for USP plotted with respect to K in blue. Overall accuracy OA% for CSP in purple dashed.

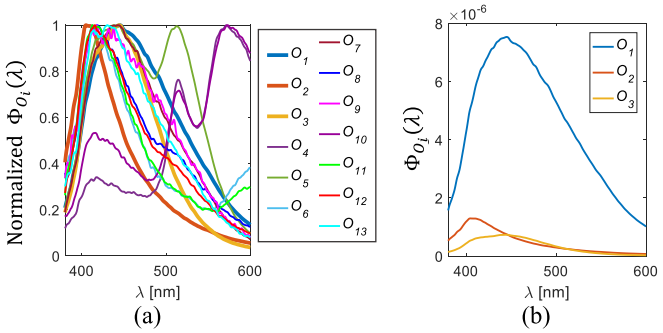


Fig. 13. (a) Fluorescence spectral signatures of the 13 object materials included in the DB employed for recognition. To increase the readability, the spectra are normalized within [0, 1]. (b) Fluorescence spectral signatures of the object materials retained for testing and submerged in the tank. These objects correspond to $\{O_1, O_2, O_3\}$ of the DB [highlighted in bold in (a)].

will be the preferred subspace projection approach employed within ARTEMIdE.

B. Experiments With Real Data

1) Experimental Design:

a) Real data collection: Real data from submerged objects were collected in laboratory by means of an artificial water column and a prototypal fluorescence LIDAR developed at the "Nello Carrara" Institute of Applied Physics–National Research Council (IFAC-CNR) Laboratory, Sesto Fiorentino, Florence, Italy. The characteristics of such LIDAR system are listed in Table VI.

TABLE VI
PROTOTYPAL LIDAR SYSTEM PARAMETERS FOR REAL DATA EXPERIMENTS

Excitation wavelength [nm]	355
Pulse energy [mJ]	8
Pulse FWHM [ns]	5
TX laser beam div. [mrad]	3
Emission efficiency	0.6
Telescope diameter [cm]	25
Receiver optical efficiency	0.5
Receiver field of view [sr]	$1.1\ 10^{-3}$
RX spectral range [nm]	[380, 600]
RX spectral resolution [nm]	4
Detector size [μm]	24
Detector focal length [mm]	300
Detector quantum efficiency	0.75
Gate duration [ns]	5
Gating delay [ns]	1

TABLE VII
DB OBJECT MATERIALS FOR REAL DATA EXPERIMENTS

O_1	Fiberglass
O_2	Neoprene
O_3	Blue Tin
O_4	Alpine Green Aluminum
O_5	Orange Aluminum
O_6	Opaque Orange Aluminum
O_7	Dark Green Aluminum
O_8	Greyish Green Aluminum
O_9	Green Aluminum
O_{10}	Orange Fiberglass
O_{11}	Opaque Orange Fiberglass
O_{12}	Greyish Green Fiberglass
O_{13}	Green Fiberglass

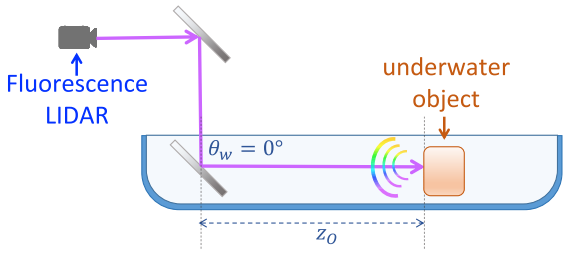


Fig. 14. Laboratory acquisition geometry.

b) Real data DB: A spectral DB was built by recording the fluorescence spectral signatures of $N = 13$ different object materials. The resulting spectra, normalized within [0, 1], are shown in Fig. 13(a), and the object materials are described in Table VII. Considerations similar to those made for the DB used for the synthetic

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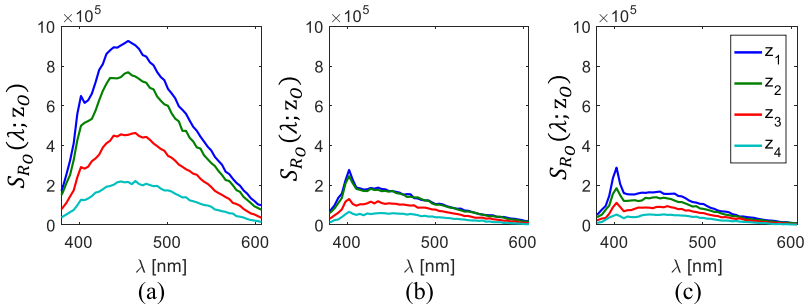


Fig. 15. Received test signals from the submerged objects at the four tested depths. (a) O_1. (b) O_2. (c) O_3.

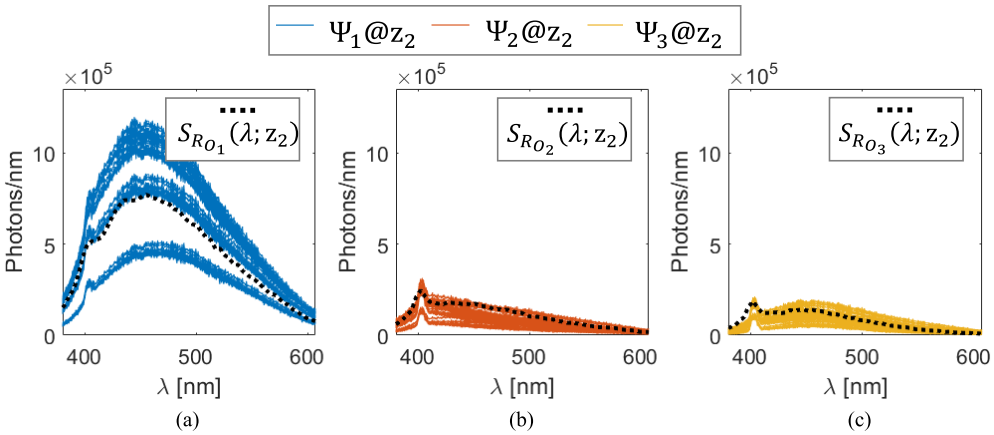


Fig. 16. Signature spaces (a) Ψ_1 (for O_1), (b) Ψ_2 (for O_2), and (c) Ψ_3 (for O_3). The signature spaces were obtained with the combination of the parameters in Table VIII. The acquired signals for the three submerged objects were superimposed in dashed black for convenience. All signals displayed in this figure were obtained for z_2 depth.

TABLE VIII
ENVIRONMENTAL PARAMETERS EMPLOYED FOR SIGNATURE SPACE GENERATION FOR REAL DATA EXPERIMENTS

$C\ [mg/m^3]$	$X\ [m^{-1}]$	$Y\ [m^{-1}]$	HA type	FA type	$Y_{HA}\ [g/m^3]$	$Y_{FA}\ [g/m^3]$
$12.30 \cdot 10^{-2}$	$5.60 \cdot 10^{-2}$	$1.61 \cdot 10^{-2}$	HA10	FA11	$9.00 \cdot 10^{-3}$	$5.80 \cdot 10^{-2}$
$39.40 \cdot 10^{-2}$	$1.43 \cdot 10^{-2}$	$4.11 \cdot 10^{-2}$	HA10	FA9	$1.40 \cdot 10^{-2}$	$2.34 \cdot 10^{-1}$
$4.00 \cdot 10^{-2}$	$2.51 \cdot 10^{-2}$	$7.20 \cdot 10^{-3}$	HA8	FA8	$1.00 \cdot 10^{-2}$	$1.39 \cdot 10^{-1}$
$25.10 \cdot 10^{-2}$	$50.10 \cdot 10^{-2}$	$15.09 \cdot 10^{-2}$	HA4	FA8	$2.90 \cdot 10^{-2}$	$2.23 \cdot 10^{-1}$

data experiments can be made for these object materials as well. Among these 13 objects, the objects $\{O_1, O_2, O_3\}$ were retained for testing. Their nonnormalized fluorescence spectral signatures are shown in Fig. 13(b).

A 7-m-long tank, serving as artificial water column, was set up in laboratory and filled with tap water. The LIDAR system was arranged so that the laser beam propagated horizontally within the water by means of fixed folding mirrors. The laboratory acquisition geometry is depicted in Fig. 14.

The three tested objects were submerged in the tank at different distances (depths), and the corresponding return signals were recorded. Specifically, four depths were considered for z_O, namely, $\{z_i\}_{i=1}^4 = \{1, 1.5, 3, 4\}$ m. Although these values and the tank length might seem short, it is worth pointing out that the LIDAR was a very low-power system constructed *ad hoc* as a laboratory demonstrator. Namely, it employed 8 mJ of pulse energy with respect to the 50 mJ of the previous experiment. Thus, the experiments can be considered totally relevant.

The test signals received from the objects when submerged at the four tested depths are shown in Fig. 15. As expected, the strongly fluorescent O_1 originated stronger return signals than the other objects.

b) Signature space generation: A number of 64 environmental conditions were considered by allowing for four configurations for the triplets (C, X, Y), four configurations for the HA and FA acid type pairs, and four configurations for the $(Y_{\text{HA}}, Y_{\text{FA}})$ pairs. The specific values employed are listed in Table VIII [28], [29]. Such values mostly refer to typical conditions of Case 1 waters, although some parameters resembling Case 2 waters were considered as well. As in the previous example, noise was varied with $L|\text{dB} = \{20, 30, 50\}$, thus obtaining signature spaces composed by $D = 64 * 3 = 192$ spectra. The signature spaces, evaluated at

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TABLE IX
STATE-OF-THE-ART COMPARISON ALGORITHMS

PHYSICS BASED METHODOLOGY	SIMILARITY MEASURES FOR SPECTRAL MATCHING	MULTIPLE HYPOTHES TESTING
Water-only Ancillary Data-driven Fluorescence Retrieval (WADFR)	**Spectral Angle Measure (SAM)** $s_{SAM}(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) = \frac{\hat{\boldsymbol{\Phi}}_O^T \cdot \boldsymbol{\Phi}_{O_i}}{\|\hat{\boldsymbol{\Phi}}_O\| \cdot \|\boldsymbol{\Phi}_{O_i}\|}$ *Invariant to scaling factors*	$H_k: \begin{cases} k = \arg max_{i=1,2,\ldots,N} \{s(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i})\} \\ s(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_k}) > \zeta_{H_0} \end{cases}$ $s(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) = \begin{cases} s_{SAM}(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) \\ s_{SCM}(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) \\ s_{DTW}(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) \end{cases}$
	Spectral Correlation Measure (SCM) $s_{SCM}(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) = \frac{d \cdot \hat{\boldsymbol{\Phi}}_O^T \cdot \boldsymbol{\Phi}_{O_i} - d^2 \cdot \langle \hat{\boldsymbol{\Phi}}_O \rangle \cdot \langle \boldsymbol{\Phi}_{O_i} \rangle}{\sqrt{\left[d \cdot \|\hat{\boldsymbol{\Phi}}_O\|^2 - d^2 \cdot \langle \hat{\boldsymbol{\Phi}}_O \rangle^2\right] \cdot \left[d \cdot \|\boldsymbol{\Phi}_{O_i}\|^2 - d^2 \cdot \langle \boldsymbol{\Phi}_{O_i} \rangle^2\right]}}$ *Invariant to scaling factors and to variations of the spectral mean*	
	Dynamic Time Warping (DTW) based Measure $s_{DTW}(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i}) = -\, d_{DTW}\left(\hat{\boldsymbol{\Phi}}_O^{(n)}, \boldsymbol{\Phi}_{O_i}^{(n)}; w\right)$ *Invariant to scaling factors and robust to non-linear distortions in the wavelength domain*	

$\hat{\boldsymbol{\Phi}}_O$	Retrieved spectral signature of the submerged object	$(\ )^{(n)}$	Vector normalized to its maximum element
$(\ )^T$	Transpose operator	d_{DTW}	DTW distance
$\|\ \|$	L_2 norm	w	Warping window half size
$\langle\ \rangle$	Mean value of the vector elements		

one of the tested depths (i.e., $z_O = z_2$), are shown in Fig. 16(a)–(c) for the three objects. The corresponding test signals received by the LIDAR system are superimposed in dashed black for comparison and reveal to be well represented by the corresponding signature space.

c) ARTEMIdE settings and algorithmic comparison: ARTEMIdE was applied by employing the CSP as subspace projection approach.

As regards the comparison with state-of-the-art methods, ARTEMIdE was compared with the automated underwater object recognition method developed in [4] and mentioned earlier (see also Table IX). The method first performs a physics-based retrieval of the submerged object fluorescence signature (up to a scaling factor) on the basis of water-only ancillary data acquired in a water column free of any object. This phase will be hereinafter denoted by water-only ancillary data-driven fluorescence retrieval (WADFR). After an estimate $\hat{\boldsymbol{\Phi}}_O$ of the submerged object fluorescence signature is retrieved, a spectral matching procedure is carried out with respect to the fluorescence signatures of the DB object materials. The DB object O_k providing the highest spectral similarity $s(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_k})$ with the retrieved signature (and higher than a suitable threshold) is selected as candidate for recognition. Several spectral similarity measurements may be employed. As in [4], here, we employ the three similarity measures summarized in Table IX (which also reports formulas, notation thereof, and main properties). The spectral angle measure (SAM) [31] evaluates the cosine of the angle subtended in $\mathbb{R}^d$ by $\hat{\boldsymbol{\Phi}}_O$ and the fluorescence signature of the ith DB object material $\boldsymbol{\Phi}_{O_i}$. The spectral correlation measure (SCM) [31] evaluates the sample correlation coefficient between $\hat{\boldsymbol{\Phi}}_O$ and Φ_{O_i}. The dynamic time warping (DTW)-based similarity measure [4] is based on the DTW distance [32] that, by warping nonlinearly $\hat{\boldsymbol{\Phi}}_O$ and $\boldsymbol{\Phi}_{O_i}$, is robust to possible nonlinear variations in the wavelength domain. The readers are referred for more details to [32] and [33]. Basically, the DTW allows for one-to-many-points comparison and a $2w + 1$ warping window may be adopted to avoid pathological warping [33].

In summary, we compared ARTEMIdE to nine algorithms, obtained by applying WADFR followed by SAM, SCM, and DTW applied with $w = \{1, 3, 5, 7, 15, 20\}$ and without warping window.

d) Performance evaluation measures: In this real-data experiment, three tested objects submerged at four different depths (one realization each) were to be recognized with respect to a DB of $N = 13$ object materials. Thus, no confusion matrix derived metrics could be adopted for performance evaluation. Similarly to what employed in [4], for the given submerged object O, a ranking of candidate DB objects was provided for recognition. The ranking list was built by sorting the DB object materials according to an ascending order of the corresponding projection errors $\{\varepsilon_{\Psi_i}^{(O)}\}_{i=1}^N$ (the lower the better) for ARTEMIdE and to a descending order of the corresponding spectral similarity measures $\{s(\hat{\boldsymbol{\Phi}}_O, \boldsymbol{\Phi}_{O_i})\}_{i=1}^N$ (the higher the better) for the WADFR-based algorithms. The lower the rank of the DB object truly corresponding to the submerged object O, the better is the performance. A rank equal to 1 means a correct recognition, whereas a rank equal to $m \leq N$ means that there are other $m - 1$ object materials in the DB that scored better.

2) Results Discussion: Fig. 17 shows bar graphs of the ranking values obtained by ARTEMIdE and by the comparison algorithms for the DB objects corresponding to the three tested objects (i.e., $\{O_1, O_2, O_3\}$). The four values reported for each

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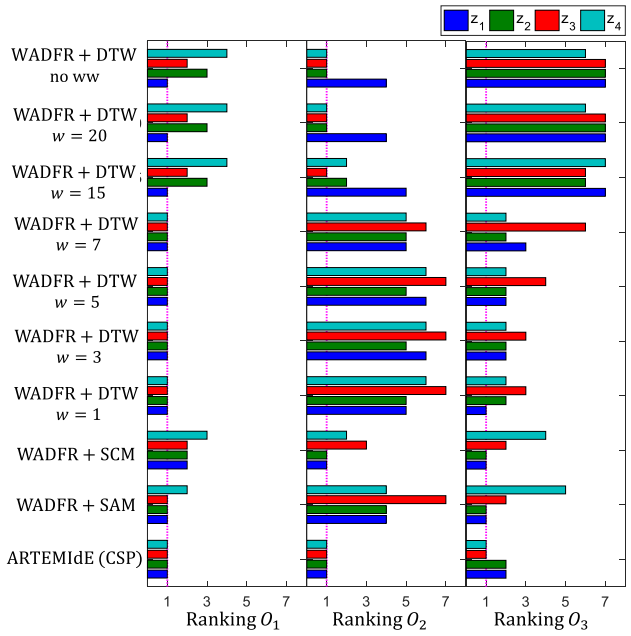


Fig. 17. Bar graphs of ranking values obtained by ARTEMIdE and by the comparison algorithms for the DB objects corresponding to the three tested objects (i.e., $\{O_1, O_2, O_3\}$). Four ranking values are reported for each combination of object and algorithm, each for one of the four tested depths. The lower the ranking value, the better the performance. The magenta dashed line highlights ranking values equal to 1 (i.e., correct recognitions).

combination of object and algorithm represent the ranking values obtained with the tested object submerged at the four different depths.

Fig. 17 shows that with ARTEMIdE, the DB objects corresponding to the tested objects ranked first in most cases. In this regard, ARTEMIdE performed similarly to the WADFR + DTW algorithm, if applied with $w \in [1, 7]$ for object O_1. In general, ARTEMIdE performed for all objects much better than all algorithms, which provided much worse ranking values. Actually, ARTEMIdE was the only algorithm that provided very good performance (nearly all correct recognitions) for all three tested objects when submerged at the four tested depths. We remind again that ARTEMIdE very good performance was obtained without making use of any additional *a priori* information or data, differently from WADFR-based algorithms, which use water-only ancillary data. Furthermore, ARTEMIdE does not require tuning any free parameters, differently from DTW application for which no theoretical criteria may be invoked.

IV. Conclusion

This article has shown that recognizing submerged objects with fluorescence LIDAR is possible even when environmental conditions are not known. Furthermore, this can be achieved without exploiting any additional *a priori* information or data, which are seldom available in operational scenarios. Specifically, an automated underwater material recognition method for fluorescence LIDAR invariant to environmental conditions has been proposed here. ARTEMIdE is based on an underwater fluorescence LIDAR signal model and exploits invariance concepts and subspace projection approaches. By generating even large signature spaces spanning the range of variation over which invariance is sought, every type of environmental variability can be accounted for. Two approaches for subspace projection have been explored. The CSP not only bypasses the difficult task of estimating the optimal subspace dimensionality, but it is also more effective at capturing the variability essential for recognition, as the extensive experimental results over synthetic data have revelaed. Experimental results over real data have shown that ARTEMIdE outperforms state-of-the-art algorithms based on a compensation procedure exploiting water-only ancillary data (WADFR) and several different similarity measures.

In the future, experiments featuring a higher number of object materials will be performed. Furthermore, invariance to acquisition geometry will also be investigated in order to extend ARTEMIdE to those applications where viewing conditions may vary during data acquisition.

ARTEMIdE will be of specific usefulness in those applications, where a system providing both range-resolved and bio-optical measurements is in demand and, at the same time, environmental conditions are hardly predictable or their contextual measurement is not feasible or difficult. One such application is, for instance, Arctic substrates monitoring [3], [35], which has been gaining more and more attention due to its relevance in climate change studies and for which fluorescence LIDAR might become the leading technology in the near future.

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